

Question	Answer	Mark
----------	--------	------

Number		
5(a)(i)	The only correct answer is B endemic <i>A is not correct because endemic is the correct term</i> <i>C is not correct because endemic is the correct term</i> <i>D is not correct because endemic is the correct term</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(a)(ii)	An answer that includes the following points: - increases the survival (of fertilised egg cells / offspring) / increase {chance / number} of egg cells hatching (1) - by {reducing predation / providing suitable conditions for development} (1)	ACCEPT increase {number of offspring / population} / prevent decrease in population accept so the egg cells {survive/ more hatch} ignore protection unqualified ACCEPT egg cells {are not eaten / can develop safely / protect from predators / are safe from other species / are safe from other fish} ACCEPT suitable conditions e.g. increased oxygen / nutrients / maintain temperature ACCEPT so the egg cells are not washed away/don't get lost	(2)

Question	Answer	Additional guidance	Mark
----------	--------	---------------------	------

Number			
5(b)(i)	An answer that includes the following point: as (sperm) length increases the (sperm swimming) speed increases / positive correlation between (sperm) length and (sperm swimming) speed (1)	ignore {linear / positive / proportional} relationship	(1)

Question Number	Answer	Mark
5(b)(ii)	The only correct answer is B correlation coefficient <i>A is not correct because chi squared test is used to test the significance of the difference between observed and expected results</i> <i>C is not correct because index of diversity is used to compare biodiversity in different habitats</i> <i>D is not correct because a students t-test is used to compare two independent groups of data to see if they are significantly different</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(b)(iii)	An answer that includes the following point: (a faster swimming speed) increases chance of fertilising an egg cell (before competitors / before death of sperm) (1)	ACCEPT increases chance of fertilisation (of egg cell) ACCEPT higher reproductive success / more likely to pass on alleles to offspring / fertilise the egg cell before competitors sperm ignore faster fertilisation ignore reaching the egg cell	(1)

Question Number	Answer	Mark
5(c)(i)	The only correct answer is A W <i>B is not correct because X is not the acrosome</i> <i>C is not correct because Y is the haploid nucleus</i> <i>D is not correct because Z is a mitochondrion</i>	(1)

Question Number	Answer	Additional guidance	Mark
5(c)(ii)	• conversion of actual length of scale into μm (1) • correct magnification given to 2 significant figures (1)	<u>Example of calculation:</u> $39 \pm 1 \text{ mm} = 39\,000 \mu\text{m}$ ALLOW answer in range of 38 000 - 40 000 μm $(39\,000 \div 60) = 650$ ALLOW 630 / 670 Allow ecf Correct answer with no working shown gains both marks	(2)

Question Number	Answer	Additional guidance	Mark
5(c)(iii)	An answer that includes three of the following points: <u>Similarities</u> • both contain (digestive / hydrolytic) enzymes (1) • both {have single membrane / membrane bound (organelles)} (1) <u>Differences (max 2)</u> • a lysosome is spherical whereas an acrosome is {curved / not} (1) • a lysosome is involved with intracellular digestion whereas an acrosome is involved with extracellular digestion (1) • a lysosome (enzyme) is involved in breaking down {cell components / microbe cells / virus} whereas an acrosome (enzyme) is involved in digesting the {outer layers of an egg cell / zona pellucida} (1)	ACCEPT both {use enzymes / involved in digestion} do not credit both are enzymes ACCEPT both surrounded by phospholipid bilayer ignore references to size ACCEPT a lysosome (enzyme) is involved in breaking down {cell components / microbe cells / virus / phagocytosis / autophagy / worn out cells / autolysis} whereas an acrosome is not ACCEPT an acrosome (enzyme) is involved in {digesting / penetrating} the {outer layers of an egg cell / zona pellucida} whereas a lysosome is not	(3)